# 6. Banking

Johan Hombert

hombert@hec.fr

# Road Map

Asset Transformation

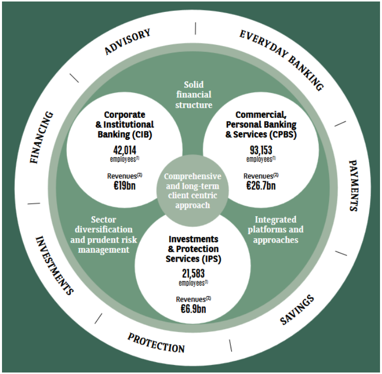
Credit Risk

Interest Rate Risk

Liquidity Risk

# Banks

- Large banks are active in
  - Commercial banking: take deposits, make loans
  - Investment banking: underwriting, M&A, trading
  - Asset management
- Ex.: BNP Paribas



# Commercial Banking

- Today: focus on commercial banking

What are the economic functions of commercial banks – why don't households lend directly to firms and to other households?

# Asset Transformation

Take deposits + Make loans = Asset transformation

# Asset Transformation

Take deposits + Make loans = Asset transformation

1. **Liquidity transformation**

    - Deposits are liquid: they are demandable
    - Loans are illiquid: banks cannot call loans; secondary markets for loans exist but are illiquid

## Asset Transformation

Take deposits + Make loans = Asset transformation

1. **Liquidity transformation**
   - Deposits are liquid: they are demandable
   - Loans are illiquid: banks cannot call loans; secondary markets for loans exist but are illiquid
2. **Maturity transformation**
   - Borrow short-term: deposits, certificates of deposits, repurchase agreements (repos), interbank loans
   - Lend long-term: business loans, mortgages, government bonds

# Asset Transformation

Take deposits + Make loans = Asset transformation

1. **Liquidity transformation**
   - Deposits are liquid: they are demandable
   - Loans are illiquid: banks cannot call loans; secondary markets for loans exist but are illiquid
2. **Maturity transformation**
   - Borrow short-term: deposits, certificates of deposits, repurchase agreements (repos), interbank loans
   - Lend long-term: business loans, mortgages, government bonds
3. **Credit risk transformation**
   - Deposits are safe
   - Loans are risky

# Asset Transformation

- Remark: Liquidity and maturity are related characteristics (that deposits are demandable make them both liquid and short-term) but different

Can you name assets that are liquid and long-term?

# Diversification

- Liquidity transformation and credit risk transformation are achieved through diversification
  - Not all depositors withdraw their deposits at the same time
  - Not all loans default
- Maturity transformation cannot be diversified away
  - Interest rate risk is systematic

# Risks

Transforming illiquid, long-term, risky assets into liquid, short-term, safe liabilities exposes banks to:

1. Liquidity risk

2. Interest rate risk

3. Credit risk

# Road Map

Asset Transformation

Credit Risk

Interest Rate Risk

Liquidity Risk

# Credit Risk

How do banks manage credit risk?

- **Limit individual risk**
  - Screening (e.g., credit scoring as studied in class 2)
  - Collateral: Loans may be secured by a specific asset (equipment, real estate, account receivables, entrepreneurs' personal assets)
  - Monitoring: Loans may include covenants (e.g., limits on debt-to-EBITDA, on dividends)
- **Diversify risk:** Hold loans in different sectors, different geographies, etc.
- **Sell risk**
  - Securitization: Sell part of the loans cash flow
  - Hedge using credit derivatives such as credit default swaps (uncommon)

Banks must hold equity to absorb losses due to remaining credit risk

# Road Map

# Interest Rate Risk

By borrowing short-term and lending long-term (at fixed rate), banks:

- Earn positive net interest income because long rates are usually higher than short rates (i.e., the **term premium** is positive)
- Are exposed to the risk of an increase in interest rate

# Interest Rate Risk: Example

- Deposits 100: demandable, interest rate 1%/year
- Loans 100: maturity 5 years, interest rate 2%/year (fixed rate)
- Cash flow if no change in interest rates (ignoring compounding)

| | year 0 | year 5 |
|---|---|---|
| deposits | 100 | –105 |
| loans | –100 | 110 |
| total | | +5 |

# Interest Rate Risk: Example

- Deposits 100: demandable, interest rate 1%/year
- Loans 100: maturity 5 years, interest rate 2%/year (fixed rate)
- Cash flow if no change in interest rates (ignoring compounding)

| | year 0 | year 5 |
|---|---|---|
| deposits | 100 | –105 |
| loans | –100 | 110 |
| total | | +5 |

- If the yield curve shifts up by 2%

| | year 0 | year 5 |
|---|---|---|
| deposits | 100 | –115 |
| loans | –100 | 110 |
| total | | –5 |

# Duration

## Refresher from Financial Markets Course

- The loss of 10 can be also calculated using the duration formula:

$$(\Delta\text{Value}) \simeq -\,(\Delta\text{Interest rate}) \times (\text{Duration in years})$$

- Yield curve shifts up by 2%

- Duration of assets: 5 years
  Duration of liabilities: 0

$\Rightarrow$ $\Delta$Value of assets $\simeq -2\% \times 5 = -10\%$
$\Delta$Value of liabilities $\simeq -2\% \times 0 = 0$

# Interest Rate Risk

How do banks manage interest rate risk?

- **Asset Liability Management**: Manage the mismatch between assets duration and liabilities duration
  - Borrow longer term
  - Lend shorter term
  - Lend at variable rate
  - Hedge using interest rate derivatives
- But reducing the duration mismatch implies giving up on the term premium

Banks hold equity to absorb losses due to remaining interest rate risk

# Interest Rate Risk

Reuters

## SVB is largest bank failure since 2008 financial crisis

March 11, 2023 6:15 AM GMT+1 · Updated March 11, 2023

BBC

## Silicon Valley Bank: Rising interest rates will uncover more ticking bombs

12 March 2023

# U.S. Banking System, March 2023

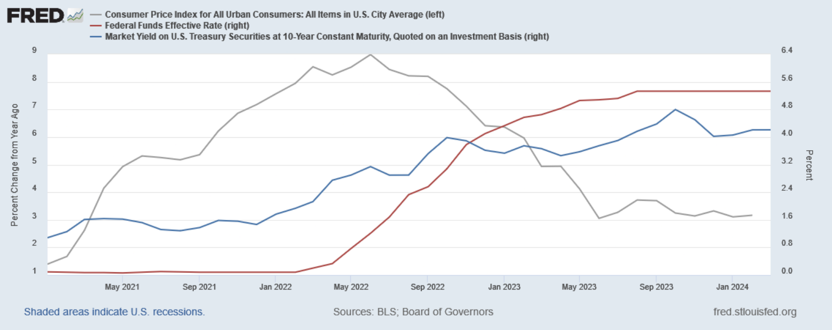
**Bank Asset Composition – 2022Q1**

| | |
|---|---|
| Total Asset $ | 24T |
| Number of Banks | 4844 |
| (Percentage of Asset) | |
| Cash | 14.1 |
| Security | 25.2 |
| Treasury | 6.1 |
| RMBS | 12.1 |
| CMBS | 2.3 |
| ABS | 2.7 |
| Other Security | 2.1 |
| Total Loan | 46.6 |
| Real Estate Loan | 21.9 |
| Residential Mortgage | 10.6 |
| Commercial Mortgage | 2.2 |
| Other Real Estate Loan | 9.1 |
| Agricultural Loan | 0.3 |
| Commercial & Industrial Loan | 9 |
| Consumer Loan | 7.7 |
| Loan to Non-Depository | 2.8 |
| Fed Funds Sold | 0.1 |
| Reverse Repo | 1.2 |

**Bank Liability Composition – 2022Q1**

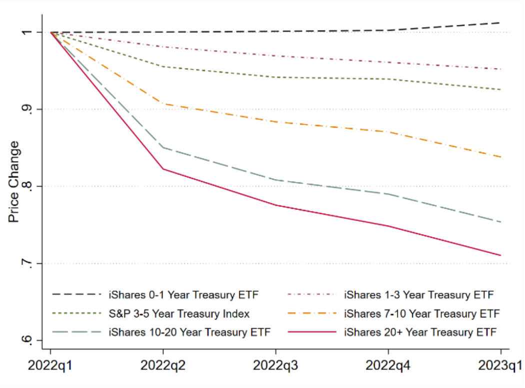
| | |
|---|---|
| Total Liability | 90.5 |
| Domestic Deposit | 76.6 |
| Insured Deposit | 41.1 |
| Uninsured Deposit | 37.4 |
| Uninsured Time Deposits | 1.8 |
| Uninsured Long-Term Time Deposits | 0.4 |
| Uninsured Short-Term Time Deposits | 1.3 |
| Foreign Deposit | 6.5 |
| Fed Fund Purchase | 0.1 |
| Repo | 0.6 |
| Other Liability | 2.3 |
| Total Equity | 9.5 |
| Common Stock | 0.2 |
| Preferred Stock | 0.1 |
| Retained Earning | 4 |

# U.S. March 2023



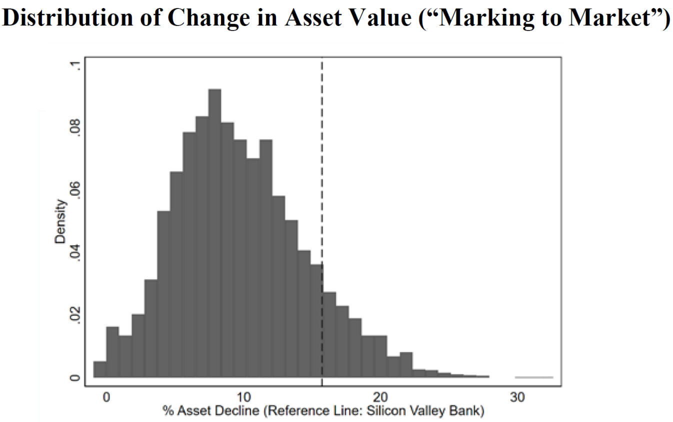

Source: https://fred.stlouisfed.org/

# U.S. Banking System, March 2023




(c) Treasury by Maturity

# U.S. Banking System, March 2023




Source: Jiang, Matvos, Piskorski and Seru, "Monetary Tightening and U.S. Bank Fragility in 2023: Mark-to-Market Losses and Uninsured Depositor Runs?" Journal of Financial Economics, 2024 [pdf]

# Interest Rate Risk

🤔 Most U.S. banks did not fail in 2023 despite large losses, sometimes larger than equity. Why not?

# Interest Rate Risk: Sticky Deposit Rates

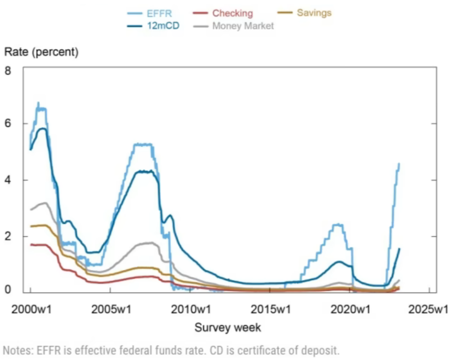
Most U.S. banks did not fail in 2023 despite large losses, sometimes larger than equity. Why not?

- Deposits do not fully reprice
  - Back to our example: deposit rate 1%, 5-year loan fixed rate 2%
  - Suppose yield curve shifts up by 2%

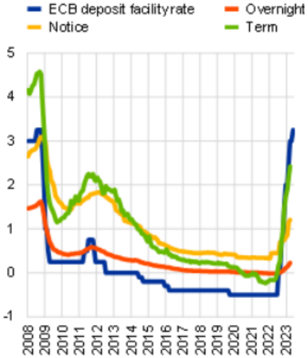
  ... but deposit rate increases by 1% only

| | year 0 | year 5 |
|---|---|---|
| deposits | 100 | ~~−115~~ −110 |
| loans | −100 | 110 |
| total | | ~~−5~~ 0 |

# Interest Rate Risk: Sticky Deposit Rates




United States




Euro Area

- The **deposit beta**, defined as the sensitivity of deposits rates to changes in the short rate, is far below one
- Due to depositors "sleepiness" giving banks market power in deposits rate setting

# Road Map

# Liquidity Risk — Game

- Players
  - 1 bank: Johan
  - N short-term creditors: You and the N-1 other students in the room
- You hold 100 worth of debt on the bank expiring now. Choose btw:
  - Roll-over at 2% interest rate → the bank owes you 102 at maturity
  - Stop lending → the bank pays you 100 now
- Bank's balance sheet:

| Assets | Liabilities |
|---|---|
| Loans: $N \times 100$ | Short-term debt: $N \times 100$ |

- Bank's assets payoff to the bank (per \$1 of book value today):
  - 1.05 at maturity if held until maturity
  - 1 now if liquidated now

# Liquidity Risk — Game

- Players
  - 1 bank: Johan
  - N short-term creditors: You and the N-1 other students in the room
- You hold 100 worth of debt on the bank expiring now. Choose btw:
  - Roll-over at 2% interest rate → the bank owes you 102 at maturity
  - Stop lending → the bank pays you 100 now
- Bank's balance sheet:

| Assets | Liabilities |
|---|---|
| Loans: $N \times 100$ | Short-term debt: $N \times 100$ |

- Bank's assets payoff to the bank (per \$1 of book value today):
  - 1.05 at maturity if held until maturity
  - 1 now if liquidated now


Decide whether to roll-over or stop lending
https://forms.gle/jXmcEkGFKT94fRnw5

# Illiquid Bank's Assets

- Now, the bank's assets are illiquid. They pay off:

  1.05 at maturity if held until maturity

  0.80 now if liquidated now

- The rest is the same

# Illiquid Bank's Assets

- Now, the bank's assets are illiquid. They pay off:

  1.05 at maturity if held until maturity

  0.80 now if liquidated now

- The rest is the same

Decide whether to roll-over or stop lending
https://forms.gle/jXmcEkGFKT94fRnw5

[spreadsheet]

# Bank Runs

- Best response depends on expectations of what other depositors do
  - If other depositors roll over $\Rightarrow$ Should roll over
  - If other depositors stop lending $\Rightarrow$ Should stop lending

$\Rightarrow$ Two possible (self-fulfilling) equilibria

  - Business as usual: all depositors roll over
  - "Bank run": all depositors stop lending

🤔 How to avoid bank runs?

# Deposit Insurance

- Deposit insurance: Government commits to repay depositors if the bank is insolvent $\Rightarrow$ Depositors have no incentives to run $\Rightarrow$ Self-fulfilling
- Costless for the government since run does not actually occur
- Deposits are usually insured up to a maximum amount (France €100k, US \$250k)

# Silicon Valley Bank, March 2023

- 90% of SVB's deposits were uninsured

# Run Risk Beyond Banks

- Run risk exists for financial intermediaries with **short-term liabilities + illiquid assets**

- Are equity mutual funds exposed to runs?

# Run Risk Beyond Banks

- Run risk exists for financial intermediaries with **short-term liabilities + illiquid assets**
- Are equity mutual funds exposed to runs?
- Are private equity funds exposed to runs?

# Run Risk Beyond Banks

- Run risk exists for financial intermediaries with **short-term liabilities + illiquid assets**
- Are equity mutual funds exposed to runs?
- Are private equity funds exposed to runs?
- High demand for private equity and private debt funds led private capital firms to create such funds for individuals with redeemable shares

FINANCIAL TIMES

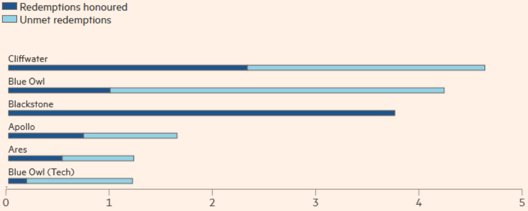
Private credit + Add to myFT

# Investors sought to pull $20bn from private credit funds in first quarter

Big groups including Apollo, Ares and Blackstone were hit with redemption requests at start of 2026

Published APR 9 2026

Quarterly redemptions fulfilled and left unmet ($bn)



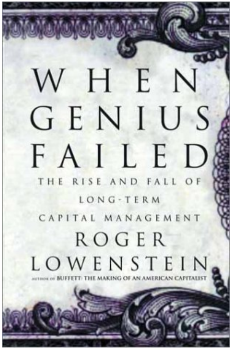

Source: Company filings, FT research

FINANCIAL TIMES

# Run Risk Beyond Banks

- Hedge funds and investment banks holding illiquid assets financed with short-term debt

LTCM failure in 1998




Lehman Brothers bankruptcy in 2008 triggering the Great Financial Crisis

# Theory of Runs

- The game you played is based on the Diamond-Dybvig model of bank runs (Nobel 2022)

## Fresh Money Injection — With Illiquid Assets

- Suppose creditors are uninsured and there is a run. We go back to the point in time just after creditors have run, before the bank liquidates assets at 80 cents on the dollar
- A deep pocket investor steps in and can provide fresh money to the bank. If this happens, the deep pocket investor's claim is junior to the existing creditors
- Please form groups of two persons: one is the bank, one is the deep pocket investor
- Goal of each party: maximize his or her profit
- The deep pocket investor's WACC is 0%
- You can only do financial restructuring, you cannot increase the value of assets
- Negotiations open!

# Fresh Money Injection — With Impaired Assets

- Now, the bank's assets are illiquid and impaired: they pay off
  - 0.95 at maturity if held until maturity
  - 0.80 now if liquidated now
- Negotiations open!

# Banks Runs — Summary

- *Definitions:*

  A **liquid** bank can repay short-term creditors even if they don't roll over.
  An **illiquid** banks can't

  A **solvent** bank can repay creditors if they roll over until asset maturity.
  An **insolvent** bank can't

- When the bank is liquid and solvent:
  - Only equilibrium: No-run (= first round of the run game)

# Banks Runs — Summary

- When the bank is illiquid and solvent:
  - Multiple equilibria: No-run and Run (= second round of the run game)
  - Depositors sleepiness mitigates run risk. Digital banking amplifies run risk
  - Deposits insurance prevents run
  - Without deposits insurance, a solvent bank can raise new funds (= first negotiation game)
- When the bank is illiquid and insolvent:
  - Run
  - An insolvent bank can't raise new funds (= second negotiation game)
    $\Rightarrow$ default (Lehman Brothers 2008) or bailout (Silicon Valley Bank 2023)
  - Deposits insurance may delay but not avoid run and failure

Can you think of a potential problem created by deposit insurance? 🤔

## Conclusion

- 6 practice problems and 2 practice exams with solution on Blackboard
- Group assignment due on May 9
- I hope you enjoyed the class and learnt a lot!